TAME-Nav: Temporal Asymmetric Mixture-of-Experts Distillation for Agile and Versatile Robot Navigation

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***Abstract*— Agile and reliable navigation is a core capability for embodied robots navigating complex spatial constraints and dynamic interactions. However, a single policy often struggles to accommodate the distinct decision-making demands of diverse scenarios. We present TAME-Nav, an asymmetric Teacher–Student navigation framework built around a temporal soft mixture of experts (Soft-MoE). The Teacher combines clean observations with causally maintained dynamic-obstacle tracks to construct privileged representations, while the Student learns complementary navigation features from corrupted observation histories through temporal encoding, soft expert routing, and latent distillation. Procedural scenario generation and parallel task-level simulation broaden training coverage, while deployment-informed asymmetric domain randomization supports sim-to-real transfer. Across in-distribution and unseen compositional scenarios, TAME-Nav achieves higher success rates, lower collision rates, and higher navigation scores than representative baselines. On a real quadruped robot, it completes all twenty trials across local-trap escape and narrow-passage navigation with sudden dynamic obstacles, with only one minor contact.**

I. Introduction

Reliable local navigation is a prerequisite for autonomous navigation in complex environments. A local navigation policy converts onboard sensory observations and a local goal into real-time motion commands, determining whether a planned route can be executed safely and continuously [1]. A capable local navigator can also reduce the burden on upstream modules by allowing them to provide sparser subgoals at a lower decision frequency [2]. Traditional model-based planners offer interpretable motion models and geometric constraints, but their performance often depends on manually designed objectives and short-horizon environment predictions [3]. Consequently, they may become trapped near concave obstacles, oscillate in narrow passages [4], or exhibit overly conservative freezing behavior in dense, interacting crowds [5]. Learning-based methods can acquire richer interaction priors from demonstrations or environmental experience [6], yet their performance remains strongly affected by the diversity of the training distribution and the expressiveness of the observation representation [7]. Learned policies may generalize poorly to unseen cluttered or confined environments [8]. Obstacle motion must be inferred from sensory histories, while explicit object-state inputs introduce deployment-time tracking dependencies [9].

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These raise two fundamental questions:

Q1. With only local observations, how can a single policy coordinate complementary navigation skills, including safe obstacle avoidance, narrow-passage traversal, and local-trap escape?

Q2. Can such coordination translate into higher task completion rates and lower collision risk in unseen compositional scenarios and real-world environments?

To address these questions, we present TAME-Nav, an asymmetric Teacher–Student framework for agile and versatile robot navigation. The Teacher fuses clean observation histories with causal dynamic-object tracks, while a temporal Soft-MoE Student learns to match its privileged navigation representation through latent distillation. Both branches share an Actor that generates body-frame velocity commands. Using only onboard observations and local goals, the deployed Student supports obstacle avoidance, narrow-passage traversal, and local-trap recovery without explicit object tracking or global-map inputs.

We further develop a massively parallel task-level simulator with procedurally generated geometric layouts and dynamic interactions to broaden training coverage. Deployment-informed randomization models sensor degradation and velocity-response uncertainty: the Student receives corrupted observations, while the Teacher and Critic retain clean observations of the underlying simulated state.

Our evaluations demonstrate consistent gains in navigation success and safety over SEA-Nav, NavRL, and DRL-DCLP under matched training budgets. TAME-Nav improves in-distribution success by over 25 percentage points over the strongest baseline and maintains its advantage across unseen sequential and multiroom scenarios. Real-world experiments further demonstrate its ability to escape local traps and negotiate narrow passages with suddenly appearing obstacles: the quadruped completes all trials in both scenarios, with only one minor contact across the twenty trials.

In summary, our main contributions are as follows.

1) We propose TAME-Nav, an asymmetric Teacher–Student framework that transfers privileged navigation representations augmented with causal dynamic-obstacle tracks from a Teacher to a deployable temporal Soft-MoE Student, enabling a single policy to handle obstacle avoidance, constrained-space traversal, and local-trap escape.
2) We develop a massively parallel task-level simulator with procedural scene generation for diverse geometric and dynamic navigation challenges, together with

deployment-informed asymmetric randomization that corrupts Student observations while preserving clean Teacher–Critic supervision.

3) We demonstrate improved navigation success and safety across in-distribution and unseen compositional scenarios, and validate reliable local-trap escape and narrow-passage navigation with sudden dynamic obstacles on a real quadruped.

II. Related Work

A. Learning-Based Local Navigation

Local navigation methods can be broadly categorized into model-based approaches and learning-based approaches. Classical methods such as DWA and TEB search for safe control commands online using explicit kinematic constraints and manually designed cost functions [3], [10]. Although these methods are interpretable and computationally efficient, their performance in complex static–dynamic environments often depends heavily on environment modeling and parameter tuning. Reinforcement learning instead learns a direct mapping from observations, such as LiDAR measurements and goal states, to continuous velocity commands [11]. Subsequent studies have incorporated dynamic-agent modeling, velocity-obstacle priors, and temporal environment representations to improve anticipatory collision avoidance in crowded environments [12], [1]. Recent work has also explored learning-based locomotion and navigation for quadruped robots in high-speed, cluttered, and open environments [13], [14]. Nevertheless, existing methods are often limited by insufficient coverage of diverse local-navigation scenarios and the high cost of data collection in high-fidelity physics simulators.

B. Teacher–Student Reinforcement Learning

Teacher–Student reinforcement learning typically allows the Teacher to access privileged information during training and transfers the acquired capability to a Student that relies only on deployable observations, commonly through action imitation or latent-representation distillation [15], [16]. Recent studies have further shown that excessive reliance on privileged information unavailable to the Student may lead the Teacher to develop behaviors that are difficult to imitate, highlighting the need to account for the Teacher–Student information gap during Teacher training [17].

This paradigm has been widely adopted in legged locomotion, where privileged terrain information or full-state observations are used to train a Teacher, while the deployable Student operates using proprioceptive signals and observation histories [18], [19]. Wu et al. [20] further incorporate a gated MoE Student into concurrent Teacher–Student learning to improve proprioceptive locomotion across diverse terrains. In navigation, DiPCAN reconstructs privileged dynamic-environment representations from historical visual observations [21], while the recent GPTS-Nav employs graph-structured privileged information to guide a LiDAR-based navigation policy in dynamic environments [22]. These studies demonstrate the value of privileged knowledge for action generation; however, most focus on observation asymmetry within the same simulation setting and employ a unified Student representation for a relatively specific class of navigation scenarios.

Our approach exploits a lightweight task-level simulator to achieve broader coverage of complex maps and failure cases. The Teacher–Student framework transfers privileged navigation knowledge to a deployable Student, while the Student's mixture-of-experts (MoE) architecture provides a structural basis for learning scene-dependent representations across heterogeneous local-navigation situations.

III. Method

Coordinating diverse navigation behaviors within a single deployable policy remains challenging. We propose TAME-Nav, an asymmetric Teacher–Student framework centered on a temporal Soft-MoE Student. The following subsections present the navigation formulation, network architecture, procedural training environments, and asymmetric domain randomization.

A. Preliminaries

1) Local Navigation Formulation: We formulate navigation in dynamic environments as a partially observable Markov decision process (POMDP) [23]. Scenarios $\xi \sim p(\xi)$ vary in obstacle geometry, start–goal configurations, and dynamic interactions. The robot receives LiDAR measurements, the relative goal, its motion state, and the previous command, as defined in Eq. (1):

$$o_t = \left[o_t^{\text{lidar}}, o_t^{\text{goal}}, o_t^{\text{motion}}, \mathbf{u}_{t-1} \right]. \tag{1}$$

To account for motion and occlusion, the policy conditions on an H-frame history $h_t = (o_{t-H+1}, \ldots, o_t)$ and generates $\mathbf{u}_t \sim \pi_\theta(\cdot \mid h_t)$. The command $\mathbf{u}_t = [v_{x,t}, v_{y,t}, \omega_{z,t}]^\top$ specifies body-frame longitudinal velocity, lateral velocity, and yaw rate.

We learn safe and efficient goal-reaching behavior by maximizing the expected discounted return in Eq. (2):

$$J(\pi_\theta) = \mathbb{E}_{\xi,\tau} \left[\sum_{t=0}^{T-1} \gamma^t r_t \right]. \tag{2}$$

Here, τ is a policy-generated trajectory in scenario ξ, r_t is the reward, and $\gamma \in [0, 1)$ is the discount factor. The episode ends at T upon success, failure, or reaching the maximum horizon.

B. Overall Architecture

As illustrated in Fig. 1, TAME-Nav adopts an asymmetric Teacher–Student architecture with separate encoders and a shared high-level Actor. The Teacher combines clean observation histories with causal dynamic-obstacle tracks, while the temporal Soft-MoE Student infers a compatible representation solely from onboard histories. During deployment-oriented training, Student observations are delayed and corrupted, while the Teacher and Critic retain clean observations and privileged information.

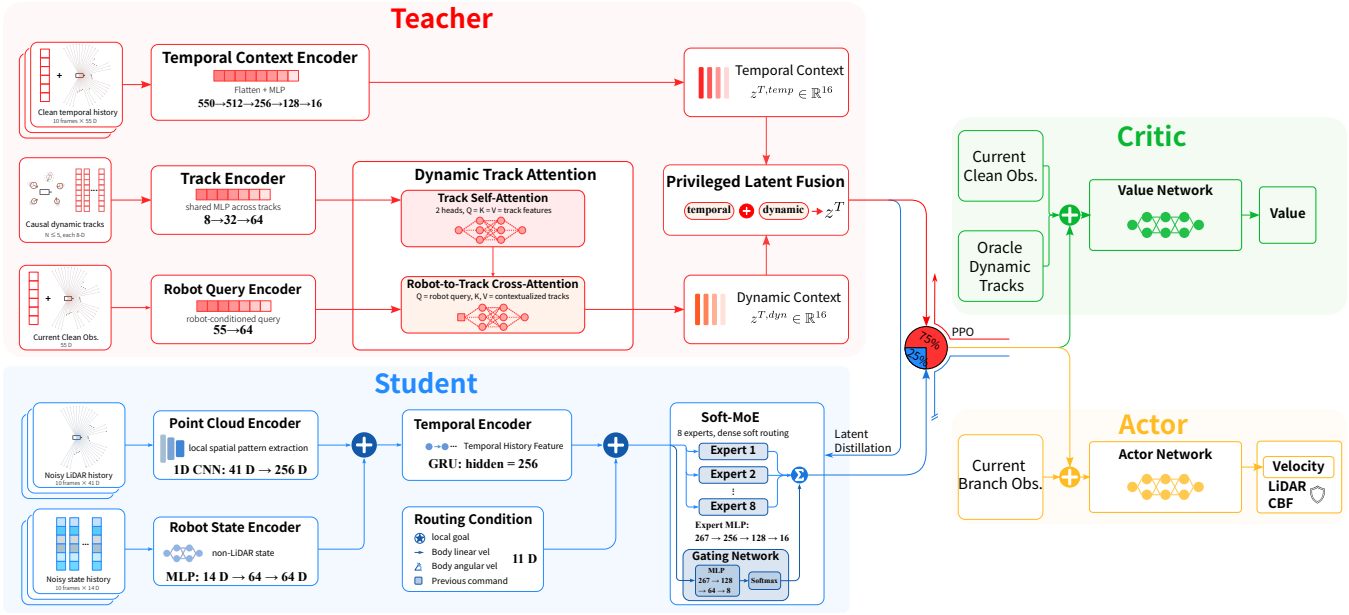


Fig. 1: Overview of TAME-Nav. The Teacher combines clean observation histories with causal dynamic tracks, while the temporal Soft-MoE Student learns from corrupted histories through latent distillation. Both share an Actor with LiDAR-based safety correction. A privileged Critic supports training; only the Student encoder, Actor, and safety module are deployed.

Given the current observation and selected latent, the Actor produces a nominal body-frame velocity mean, which a differentiable LiDAR-based CBF corrects to parameterize a Gaussian policy. Teacher and Student collect rollouts from fixed subsets of parallel environments, with 75% allocated to Teacher-controlled rollouts and 25% to Student-controlled rollouts. PPO [24] updates the Teacher encoder, shared Actor, and privileged Critic, while the Student encoder learns through latent distillation from stop-gradient Teacher representations at Student-visited states. Deployment retains only the Student encoder, shared Actor, and LiDAR-based safety module.

Track-Augmented Teacher: The Teacher combines clean observation histories with causal dynamic-obstacle tracks to construct a privileged representation for the shared Actor and Student latent distillation. Following Eq. (1), an MLP encodes a ten-frame history $\bar{\mathbf{H}}_t^T \in \mathbb{R}^{10\times 55}$ into $\mathbf{z}_t^{T,\mathrm{temp}} = E_T^{\mathrm{temp}}(\mathrm{vec}(\bar{\mathbf{H}}_t^T)) \in \mathbb{R}^{16}$.

The training environment maintains causal tracks and selects the five nearest valid obstacles, forming $\mathcal{B}_t = \{\mathbf{b}_t^i\}_{i=1}^{N_t}$ with $N_t \le 5$. Each track is defined in Eq. (3):

$$\mathbf{b}_t^i = \left[(\mathbf{p}_{t,\mathrm{rel}}^i)^\top, (\mathbf{v}_{t,\mathrm{obs}}^{i,B})^\top, \rho_t^i, \sigma_t^i, f_{t,\mathrm{miss}}^i, m_{t,\mathrm{vis}}^i\right]^\top \in \mathbb{R}^8. \tag{3}$$

Here, $\mathbf{p}_{t,\mathrm{rel}}^i$ and $\mathbf{v}_{t,\mathrm{obs}}^{i,B}$ denote robot-relative position and obstacle velocity, both expressed in body coordinates. The remaining entries represent circumscribed radius, age-dependent heuristic uncertainty, normalized observation age, and visibility. Occluded tracks are extrapolated using their latest velocity estimates until expiration; no future obstacle states are accessed.

Normalized track features pass through a shared MLP and two-head self-attention with a residual connection and LayerNorm [25]. The resulting features $\widetilde{\mathbf{E}}_t$ are padded to five entries and aggregated by two-head cross-attention, using $\mathbf{q}_t = \phi_{\mathrm{qry}}(\bar{\mathbf{o}}_t^T) \in \mathbb{R}^{64}$ as the robot-conditioned query and $\widetilde{\mathbf{E}}_t$ as keys and values. Each head has dimension 32, and padded entries are masked. LayerNorm and a linear projection produce $\mathbf{z}_t^{T,\mathrm{dyn}} \in \mathbb{R}^{16}$, which is set to zero when no valid tracks exist.

The final latent uses normalized residual fusion:

$$\mathbf{z}_t^T = \mathrm{Norm}_2\left(\mathrm{sg}(\mathbf{z}_t^{T,\mathrm{temp}}) + g_{\mathrm{dyn}}\mathbf{z}_t^{T,\mathrm{dyn}}\right). \tag{4}$$

In Eq. (4), Norm_2 denotes L2 normalization, and g_{dyn} is a learnable scalar initialized to one. Stop-gradient blocks Actor gradients to the temporal encoder, which learns through the value loss; the dynamic encoder and fusion coefficient receive policy gradients.

Temporal Soft-MoE Student: The Student combines temporal encoding with softly routed expert representations to construct a navigation latent from onboard sensing and control histories. Its observation $\widetilde{\mathbf{o}}_t^S \in \mathbb{R}^{55}$ comprises a 41-dimensional LiDAR scan and 14 non-LiDAR quantities: the body-frame goal, linear and angular velocities, previous command, and projected gravity. The tilde denotes observations potentially affected by sensing delays and corruption.

The Student maintains a history $\widetilde{\mathbf{H}}_t^S \in \mathbb{R}^{10\times 55}$. The window advances with the 10 Hz LiDAR stream, covering a nominal 0.9 s scan history. Between updates, the latest scan and associated goal observation are held, while the newest slot's proprioceptive state and previous command are refreshed at the 50 Hz control rate.

For each history slot, a three-layer 1D CNN encodes LiDAR into $\mathbf{f}_k^{\mathrm{lidar}} \in \mathbb{R}^{256}$, and a two-layer MLP encodes the remaining state into $\mathbf{f}_k^{\mathrm{state}} \in \mathbb{R}^{64}$. Their concatenation $\mathbf{f}_k = [\mathbf{f}_k^{\mathrm{lidar}}; \mathbf{f}_k^{\mathrm{state}}] \in \mathbb{R}^{320}$ is processed by a single-layer

GRU [26]. Its final hidden state summarizes the observation sequence in Eq. (5):

$$\mathbf{h}_t^S = \mathrm{GRU}\left(\mathbf{f}_1, \ldots, \mathbf{f}_{10}\right) \in \mathbb{R}^{256}. \tag{5}$$

An eight-expert Soft-MoE conditions this summary on the 11-dimensional navigation context $\boldsymbol{\chi}_t^S$, comprising the latest goal, linear and angular velocities, and previous command. Let $\mathbf{x}_t^S = [\mathbf{h}_t^S; \boldsymbol{\chi}_t^S] \in \mathbb{R}^{267}$. The expert branches share an input projection and produce separate 16-dimensional candidate representations. A learned router combines them through the dense soft aggregation [27] in Eq. (6):

$$\begin{aligned} \mathbf{w}_t &= \mathrm{softmax}\left(G_S(\mathbf{x}_t^S)\right), \\ \mathbf{z}_t^S &= \mathrm{Norm}_2\left(\sum_{i=1}^{8} w_t^i E_i(\mathbf{x}_t^S)\right) \in \mathbb{R}^{16}. \end{aligned} \tag{6}$$

Here, E_i denotes the i-th expert mapping and Norm_2 denotes L2 normalization. This formulation supports context-dependent combinations of expert features through continuous, differentiable routing.

The Student is trained by matching the Teacher's fused representation on Student-controlled rollout samples. For a minibatch $\mathcal{D}_S$, its objective is given in Eq. (7):

$$\mathcal{L}_S = \frac{1}{16|\mathcal{D}_S|} \sum_{t \in \mathcal{D}_S} \left\| \mathbf{z}_t^S - \mathrm{sg}(\mathbf{z}_t^T) \right\|_2^2 + \lambda_{\mathrm{bal}} \mathcal{L}_{\mathrm{bal}}. \tag{7}$$

The Teacher targets correspond to the same rollout states and are held fixed during Student optimization. The balancing term penalizes deviations of minibatch-mean expert weights from a uniform distribution. Only the Student encoder receives gradients from this objective; PPO gradients do not update it. The resulting representation replaces the Teacher latent at the shared Actor interface during deployment.

Shared Actor and Privileged Critic: The Teacher and Student share a high-level Actor conditioned on the current observation and the corresponding navigation latent. For branch $b \in \{T, S\}$, the shared backbone produces $\mathbf{h}_t^b = \phi_\pi([\mathbf{o}_t^b; \mathbf{z}_t^b])$. Two heads predict the nominal body-frame velocity $\boldsymbol{\mu}_t^{b,\mathrm{nom}} = \phi_{\mathrm{nav}}(\mathbf{h}_t^b)$ and a positive CBF coefficient $\alpha_t^b = \mathrm{softplus}(\phi_\alpha(\mathbf{h}_t^b))$.

We adopt SEA-Nav's differentiable LiDAR-based CBF formulation [28], which corrects the translational velocity mean while preserving the nominal yaw rate. The corrected mean parameterizes the Gaussian policy in Eq. (8):

$$\begin{aligned} \boldsymbol{\mu}_t^{b,\mathrm{cbf}} &= \mathcal{P}_{\mathrm{CBF}}\left(\boldsymbol{\mu}_t^{b,\mathrm{nom}}; \mathbf{o}_t^{b,\mathrm{lidar}}, \alpha_t^b\right), \\ \pi_\theta(\mathbf{u}_t \mid \mathbf{o}_t^b, \mathbf{z}_t^b) &= \mathcal{N}\left(\boldsymbol{\mu}_t^{b,\mathrm{cbf}}, \mathrm{diag}(\boldsymbol{\sigma}^2)\right), \end{aligned} \tag{8}$$

where $\boldsymbol{\sigma} \in \mathbb{R}^3$ is a learnable, state-independent standard deviation.

The training-only Critic receives additional privileged information [15]. An obstacle-set encoder maps the ground-truth states of up to five dynamic obstacles—body-frame relative positions, obstacle velocities, and radii—to $\mathbf{d}_t^V = E_V^{\mathrm{dyn}}(\mathcal{S}_t^{\mathrm{dyn}}, \bar{\mathbf{o}}_t^T) \in \mathbb{R}^{32}$. A normalized static cost-to-go scalar c_t^{stat} is queried from a goal-conditioned Soft-min field. Together with the current Teacher observation and its temporal latent, these features form the 104-dimensional value input in Eq. (9):

$$V_\psi(s_t) = \phi_V\left([\bar{\mathbf{o}}_t^T; c_t^{\mathrm{stat}}; \mathbf{z}_t^{T,\mathrm{temp}}; \mathbf{d}_t^V]\right). \tag{9}$$

This privileged information supports PPO value estimation without entering the Actor. Deployment retains only the Student encoder, shared Actor, and LiDAR-based safety module.

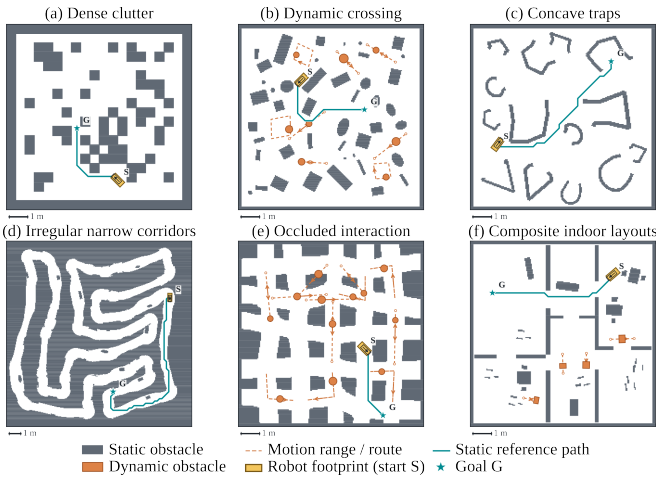


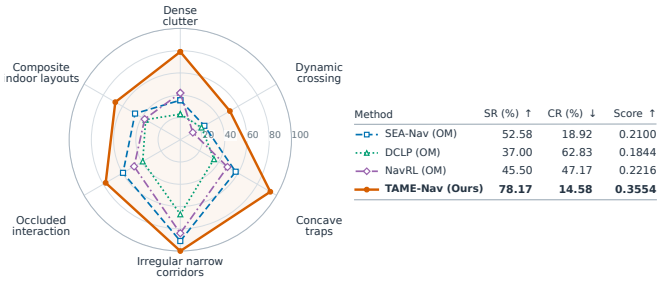
Fig. 2: Representative layouts from the six training scene families, each containing 100 distinct maps.

C. Task-Level Simulation and Procedural Scenario Generation

We train TAME-Nav in a custom GPU-vectorized 2D navigation simulator. A command-response model updates robot poses and velocities without simulating joint-level dynamics. The simulator retains footprint collisions with blocking and sliding, velocity and acceleration limits, dynamic obstacles, and LiDAR and odometry observations. At deployment, a separately trained locomotion controller executes the same velocity commands. This abstraction supports large-scale interaction and flexible scenario generation, with sensing and execution perturbations introduced during deployment-oriented training.

The training set comprises 600 layouts, evenly distributed across six scene families: dense clutter, multidirectional dynamic crossings, concave traps, irregular narrow corridors, occluded dynamic interactions, and composite indoor layouts (Fig. 2). Doorway traversal is incorporated into indoor layouts. Procedural generation varies obstacle geometry, clearance, occlusion, and dynamic motion to cover complementary navigation challenges.

Each map provides a prevalidated pool of start–goal configurations and initial headings, sampled at episode reset. Validation checks footprint clearance, static connectivity after obstacle inflation, and dynamic trajectories against map boundaries, static structures, and protected endpoint regions. Scene labels, global maps, and reference paths are excluded from Actor inputs; static cost-to-go information is used only for training-time reward shaping and value estimation.



Method	SR (%) ↑	CR (%) ↓	Score ↑
SEA-Nav (OM)	52.58	18.92	0.2100
DCLP (OM)	37.00	62.83	0.1844
NavRL (OM)	45.50	47.17	0.2216
TAME-Nav (Ours)	**78.17**	**14.58**	**0.3554**

Fig. 3: In-distribution navigation performance. (OM) denotes training on our maps. TAME-Nav achieves the highest success rate across all six scene families and the best overall SR, CR, and Score, with particularly clear gains in dense clutter and concave traps.

This design broadens scenario coverage while preserving the deployed policy's local-observation interface.

D. Asymmetric Domain Randomization

We apply deployment-informed sensing and execution randomization [29]. The Student distills Teacher latents from corrupted histories, while the Teacher and Critic retain clean observations and privileged states. Execution perturbations affect both branches; clean observations reflect the realized robot state.

LiDAR operates at $10\,\mathrm{Hz}$ with 0–$20\,\mathrm{ms}$ additional delay, 1–$2\,\mathrm{cm}$ range-noise standard deviation, and 0–3% ray dropout. Whole-scan dropout is capped at 0.3%, with the last valid scan retained. Extrinsic perturbations, 0–$40\,\mathrm{ms}$ odometry delays, and slow pose drift further corrupt Student observations.

Axis-dependent execution models capture tracking dynamics and acceleration limits. Forward-motion delay, response time constant, and gain span 0–$40\,\mathrm{ms}$, 0.08–$0.16\,\mathrm{s}$, and 0.80–0.95, respectively, with a startup dead zone of 0.22–$0.28\,\mathrm{m/s}$ and nonlinear low-speed attenuation. Previous-command observations store the final bounded and filtered commands. System parameters are sampled per episode; sensor noise and dropout are sampled at each sensor update.

IV. Experiments

A. Experimental Setup

1) Baselines: We compare against three learning-based navigation methods.

SEA-Nav [28] combines differentiable safety constraints and collision replay for efficient learning of safe, agile quadruped navigation in cluttered environments.

NavRL [9] combines reinforcement learning with velocity-obstacle-based safety for UAV navigation in dynamic environments. The authors also provide a Go2 quadruped deployment example in simulation.

DCLP [30] uses dimension-configurable representations and curriculum learning to accommodate varying robot dimensions and motion constraints without additional fine-tuning.

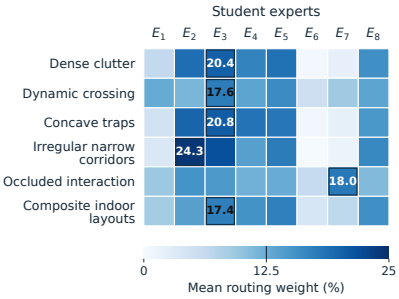


Fig. 4: Mean expert routing weights across scene families. Boxed cells indicate the largest weight in each row; 12.5% corresponds to uniform routing.

2) Training: We compare baselines trained on their original maps with methods trained from scratch on our map set; the latter baseline variants are denoted (OM). Baselines retain their original reward functions and training configurations, with aligned interaction budgets. TAME-Nav, SEA-Nav, and NavRL use equal numbers of environment interaction steps, while SAC-based DRL-DCLP adjusts its step budget according to the control period to match cumulative simulated time. Each method is evaluated using its best validation checkpoint within the allocated budget.

3) Evaluation Metrics: We report success rate (SR) and collision rate (CR). To jointly assess task completion and efficiency, we adopt the BARN navigation score used in DRL-DCLP [30], as defined in Eq. (10):

$$\text{Score} = \frac{1}{N}\sum_{i=1}^{N} S_i \frac{T_i^*}{\text{clip}(T_i,\, 2T_i^*,\, 8T_i^*)}, \tag{10}$$

where N is the number of evaluation episodes, S_i indicates success, and T_i is the actual navigation time. The reference time is $T_i^* = L_i^*/v_{\max}$, where L_i^* is the shortest path length for the episode's start–goal pair on the inflated static occupancy grid and $v_{\max} = 2.0\,\mathrm{m/s}$. The clipping operator bounds the denominator between $2T_i^*$ and $8T_i^*$. Failed episodes receive zero, and higher scores indicate better overall performance.

B. Simulation Experiments

1) In-Distribution Navigation Performance: We evaluate in-distribution generalization on 600 unseen layouts independently generated from the same procedural scene distribution as the training set.

As shown in Fig. 3, TAME-Nav achieves the highest SR across all six scene families. The largest gains occur in dense clutter and concave traps, exceeding 35 percentage points over the best baseline in each family. By contrast, corridor navigation is already comparatively strong for SEA-Nav (OM) and NavRL (OM). Overall, SEA-Nav (OM) limits collisions but incurs more non-collision failures, whereas collisions dominate the failures of DCLP (OM) and NavRL (OM). TAME-Nav reduces both failure types, and its leading Score indicates that improved task completion is accompanied by stronger time-normalized performance.

We further examine expert routing across the six scene

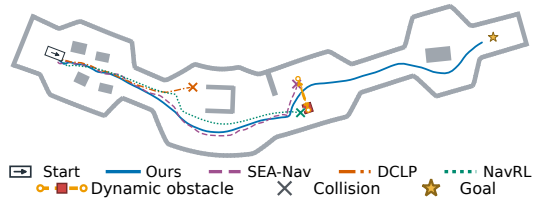


Fig. 5: An example layout from the sequential compositional test set, with trajectories of our method and the OM baselines.

TABLE I: Sequential compositional navigation performance. Our method achieves the highest SR and Score and the lowest CR. (OM) denotes training on our maps; unmarked baselines use their original maps.

Method	SR (%) ↑	CR (%) ↓	Score ↑
SEA-Nav	47.27	52.73	0.1934
NavRL	0.00	75.39	0.0000
SEA-Nav (OM)	66.80	33.01	0.2393
DCLP (OM)	59.96	40.04	0.2998
NavRL (OM)	34.96	60.94	0.1743
TAME-Nav (Ours)	**81.64**	**18.16**	**0.4068**

families in Fig. 4. E_3 receives the largest mean weight in dense clutter, dynamic crossings, concave traps, and composite indoor layouts, with E_5 receiving nearly equal weight in the latter. Narrow corridors favor E_2, whereas occluded interactions assign the greatest weight to E_7. Dense clutter and concave traps exhibit similar routing profiles, indicating substantial expert sharing. These patterns suggest that Soft-MoE adapts shared expert contributions to different navigation demands through scene-dependent routing preferences, supporting flexible specialization without a fixed correspondence between scene families and experts.

2) Sequential Compositional Navigation: We construct 64 randomized test layouts, with one example shown in Fig. 5. Each layout combines dense clutter, narrow doorways, dead-end pockets, occluded dynamic crossings, and constrained goal approaches along a single channel without external bypasses. Geometry and obstacle motion are randomized, and all layouts pass reachability checks. The 32 forward and 32 reverse tasks, each with eight matched dynamic-phase configurations, yield 512 episodes. All methods use identical initial conditions and deterministic mean actions.

We compare baselines trained on their original maps and ours. For DRL-DCLP, whose complete training code and maps are not publicly available, we report only our reproduction trained on our maps.

As shown in Table I, TAME-Nav achieves the highest SR and Score with the lowest CR. Collisions account for nearly all failures of SEA-Nav (OM) and DCLP (OM), and dominate those of NavRL (OM); the latter two incur more than twice our CR. These results highlight the difficulty of maintaining safe progress across successive challenges and support TAME-Nav's more reliable completion of composite tasks. Original NavRL records no successes, but its adapted aerial-to-ground sensing and control interfaces limit conclusions about native-system performance.

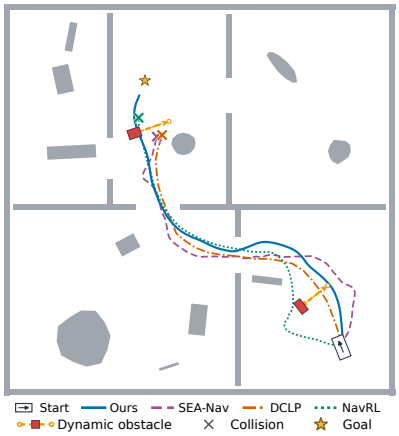


Fig. 6: An example multi-room layout with paired trajectories of our method and the OM baselines. Dynamic obstacles are shown at their initial positions together with their motion ranges; each robot trajectory spans its full episode.

TABLE II: Compositional multi-room navigation performance. Our method achieves the highest success rate and Score and the lowest collision rate in unseen multi-room environments.

Method	SR (%) ↑	CR (%) ↓	Score ↑
SEA-Nav	48.24	37.70	0.2234
NavRL	3.91	42.97	0.0195
SEA-Nav (OM)	66.41	22.85	0.2535
DCLP (OM)	60.16	35.16	0.2951
NavRL (OM)	67.58	15.62	0.3115
TAME-Nav (Ours)	**75.20**	**13.48**	**0.3448**

3) Compositional Multi-Room Navigation: We construct a compositional multi-room test set comprising 64 unseen layouts, one of which is shown in Fig. 6. Each layout contains 4–5 connected rooms, doorways 1.10–1.45 m wide, varied furniture, partially enclosed structures, and 1–2 dynamic obstacles. Room partitions, furniture arrangements, start–goal pairs, and dynamic motion parameters are randomized, and all layouts pass reachability and dynamic-trajectory validity checks. Direct start-to-goal paths are obstructed, requiring policies to combine cross-room traversal, occlusion handling, and dynamic obstacle avoidance.

We use the same models and paired evaluation protocol as in the preceding experiment, with eight dynamic-phase configurations per map and 512 episodes in total. All methods use identical start–goal pairs and dynamic initial conditions, execute deterministic mean actions, and share a 0.5 m goal threshold and a 60 s timeout.

The multiroom results in Table II show that TAME-Nav leads in SR and Score while maintaining the lowest CR. NavRL (OM) remains the closest baseline, whereas DCLP (OM) incurs more than twice our collision rate. The example trajectories highlight failures during dynamic-obstacle encounters: all three OM baselines collide, while TAME-Nav negotiates both interaction regions and reaches the goal. This case illustrates how TAME-Nav combines cross-room traversal with dynamic avoidance to sustain progress toward the goal.



Fig. 7: Real-world local-trap scenario and illustrative trajectories. Our method temporarily deviates from the goal direction to bypass the concave barrier and reach the goal.

TABLE III: Real-world local-trap navigation over ten trials per method. Our method completes all trials with only one minor contact in total, whereas no baseline exceeds 50% success. Collision counts are averaged per trial. One minor collision is allowed; a second collision, obstacle toppling, or persistent entrapment constitutes failure.

Method	SR(%)↑	Collisions/trial↓
SEA-Nav	0	0.30
NavRL	0	1.00
SEA-Nav (OM)	0	**0.00**
DCLP (OM)	0	0.20
NavRL (OM)	50	0.80
TAME-Nav (Ours)	**100**	0.10

C. Real-World Experiments

We conduct real-world experiments on a Unitree Go2 equipped with a Livox MID-360 LiDAR and an onboard NVIDIA Jetson Orin. The high-level navigation policy issues velocity commands at 50 Hz, while LiDAR observations are updated at 10 Hz.

1) Real-World Navigation in a Local-Trap Scenario: We construct a local-trap scenario comprising a concave barrier and surrounding obstacles, as shown in Fig. 7. The barrier blocks the direct start-to-goal route, requiring the robot to temporarily deviate from the goal direction and escape the trap. Each method is evaluated over ten trials. A trial permits one minor collision that does not topple an obstacle; a second collision, any collision that topples an obstacle, or persistent entrapment without progress is counted as failure.

As shown in Table III, our method completes all ten trials with only one minor collision. NavRL (OM) achieves 50% SR with 0.8 collisions per trial; some escapes occur when large-amplitude oscillations reorient the robot toward a traversable direction. All remaining baselines fail. Notably, SEA-Nav (OM) avoids collisions but remains trapped, highlighting the distinction between immediate collision avoidance and effective trap recovery. Despite training on our multi-scenario maps, the OM baselines do not achieve reliable trap escape in this real-world setting.

2) Unexpected Dynamic Obstacles in Narrow Passages: As shown in Fig. 8, we construct a winding narrow passage

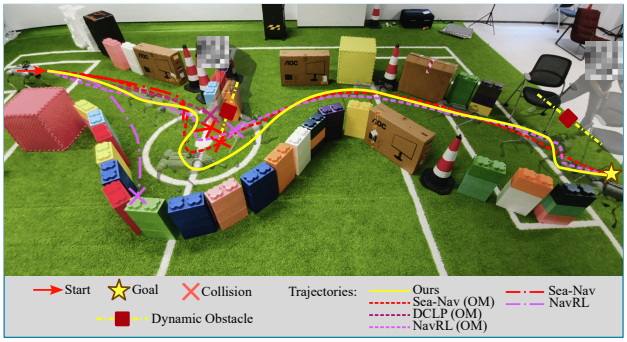


Fig. 8: Narrow-passage scenario with unexpected dynamic obstacles and illustrative trajectories. Obstacles abruptly enter the robot's path midway through the passage and near the goal. Our method negotiates both encounters and reaches the goal.

TABLE IV: Real-world dynamic obstacle avoidance in narrow passages over ten trials per method. Our method completes all trials without collisions, while successful baselines incur repeated contacts. Collision counts are averaged per trial.

Method	SR(%)↑	Collisions/trial↓
SEA-Nav	80	1.00
NavRL	0	1.00
SEA-Nav (OM)	90	0.70
DCLP (OM)	80	0.70
NavRL (OM)	90	0.70
TAME-Nav (Ours)	**100**	**0.00**

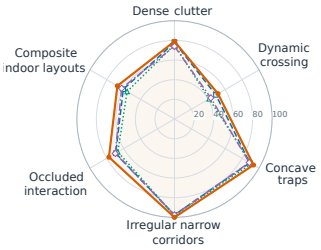
with dynamic obstacles abruptly entering the robot's path midway and near the goal. The task requires passage following, reactive avoidance, and goal approach within limited maneuvering space. Each method is evaluated over ten trials using the same collision and failure criteria as the preceding experiment.

As shown in Table IV, our method completes all ten trials without collision. SEA-Nav (OM) and NavRL (OM) achieve 90% SR, versus 80% for DCLP (OM); all three average 0.7 collisions per trial. Original SEA-Nav achieves 80% SR with one collision per trial. Original NavRL records no successes, although its aerial-to-ground transfer limits direct comparison. This result demonstrates TAME-Nav's ability to combine constrained-space traversal with collision-free avoidance of both unexpected dynamic obstacles.

D. Ablation Study

To assess the contributions of individual architectural components, we compare the full model with three ablated variants. All variants are evaluated using the same 600-map in-distribution test set and evaluation protocol as in Sec. IV-B.1.

- **w/o MoE:** Replaces the eight-expert Soft-MoE with a single parameter-matched MLP, while retaining the same temporal encoder and latent dimension.
- **w/o Teacher Tracks:** Removes the Teacher's causal dynamic-track branch, retaining its observation history.
- **w/o Temporal History:** Replaces the history with ten copies of the current observation, preserving input



Variant	SR (%) ↑	CR (%) ↓	Score ↑
Full	**78.17**	**14.58**	**0.3554**
w/o MoE	73.92	16.92	0.3409
w/o T-tracks	71.42	19.17	0.3248
w/o history	72.58	17.42	0.3315

Fig. 9: Ablation results on the in-distribution test set. All ablations degrade performance, with removing Teacher tracks causing the largest drop in success rate.

dimensions and encoder architecture.

As shown in Fig. 9, the full model achieves the best overall performance across the six scene families, with particularly clear advantages in dynamic crossing, occluded interaction, and composite indoor layouts, while all variants perform similarly in irregular narrow corridors. The w/o MoE variant reduces SR by 4.25 percentage points, while removing Teacher tracks or temporal history causes larger drops of 6.75 and 5.59 points, respectively. These results support the complementary roles of multi-expert representations, privileged dynamic supervision, and temporal information.

V. Conclusion

We presented TAME-Nav, an asymmetric Teacher–Student framework that distills track-augmented representations into a deployable temporal Soft-MoE policy. Procedural scenario generation and deployment-informed domain randomization support broad training coverage and sim-to-real transfer. Benchmark evaluations demonstrate improved navigation success, safety, and generalization, while real-quadruped experiments validate local-trap escape and narrow-passage traversal with sudden dynamic obstacles.

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